Introduction to LaTeX

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Outline

1. Introduction
2. Basic Typesetting
3. Advanced Typesetting
4. References

What is LaTeX ?

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- LaTeX is most often used to produce technical or scientific documents, but it can be used for almost any form of publishing

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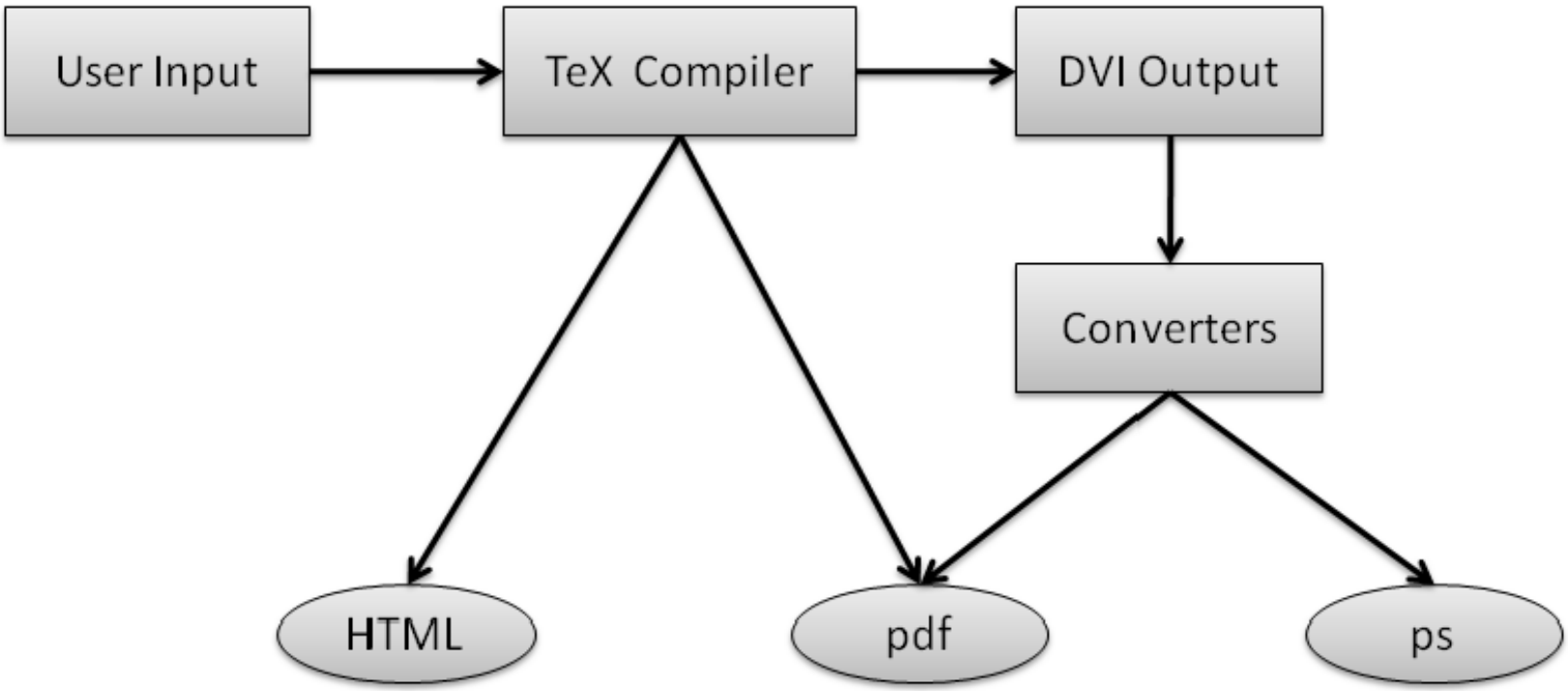
- Freely available

Basic LaTeX Work Flow I



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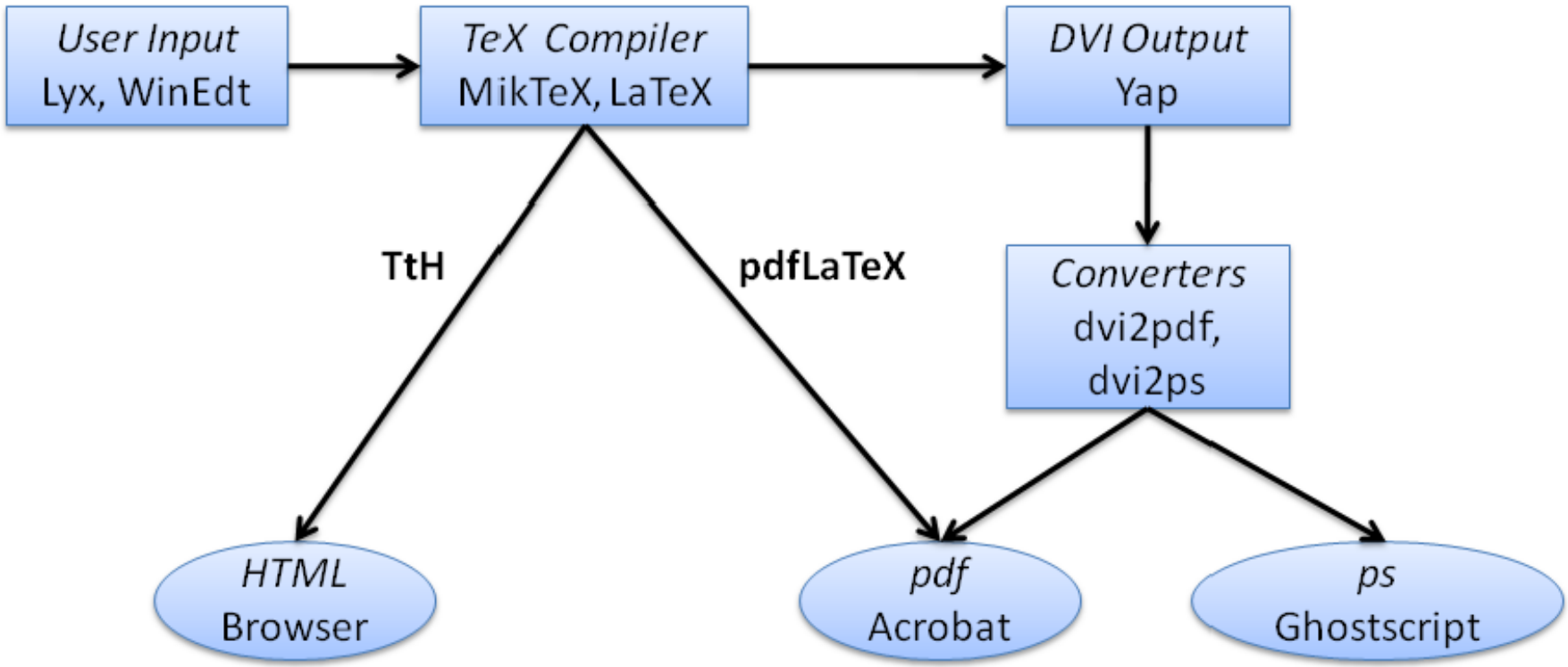
- LaTeX Output Viewer
  - PDF: xpdf, Foxit, Adobe Reader
  - PS: Ghostscript, GhostView
  - HTML: Web Browser

Basic LaTeX Work Flow II



Hello World in LaTeX

Example

```
\documentclass[12pt,letterpaper]{article}
%include packages here
%\usepackage{package}
\begin{document}
Hello world!
\end{document}
```

Basic Formatting in LaTeX

- **Bold Text**: \textbf{Bold Text }
- *Italic Text*: \emph{Italic Text }
- Spacing:
  - Many spaces = one space
  - Use \\for newline
  - Hit return *twice* for a new paragraph
  - \newpage
- Comments: % . . . your comments here . . .
- Reserved Symbols: **# $ % ^ & _ { } ~ **

Mathematical Equations in LaTeX

- Use \$ . . . \$ or \begin{math} . . . \end{math} for include mathematical symbols, equations, etc.
- Subscript and superscripts — x^2: x^2 and x_2: x_2
- Fractions— \frac{a}{b} or \$a \over b\$: $\frac{a}{b}$
- Radical— \sqrt{x}: $\sqrt{x}$
- Many more symbols and operators are available

Images and Figures in LaTeX

- Include .eps (postscript) images in LaTeX for **dvi** output
- Include .jpg, .png, and .gif images in LaTeX for **pdf** output
- Software: **gimp**: for converting images, **DIA** and **PowerPoint**: for drawing figures
- graphicx package is required: `\usepackage{graphicx}`

Example

```
\begin{figure}
\includegraphics[height=50%,width=50%]{filename.eps}
\end{figure}
```

Making Tables in LaTeX Using Excel

- Download Excel2LaTeX
  - Available for download at http://www.ctan.org/tex-archive/support/excel2latex/
- Extract the files and double click on the Excel Add-In file (Excel2LaTeX.xla)
- Create the table in Excel which you want in your LaTeX document and highlight the cells
- Click on the Macro, an error might occur, just click "End"
  - Most likely you will want to un-check "Create table environment" and "Booktabs-style formatting"
- Copy the table to your LaTeX file!

Bibliography in LaTeX I

- Bibliography information is stored in a *.bib file, in Bibtex format
- To include and reference Bibliography:
  - Set referencing style \bibliographystyle{plain}
  - Create reference section by \bibliography{bibfile with *no* extension}
  - Cite reference inside the text by \cite{*bibliography item*}

Bibliography in LaTeX II

- @book{Come95,
  author="D. E. Comer",
  title={Internetworking with TCP/IP: Principles, Protocols and Architecture},
  publisher="Prentice-Hall",
  year=1995,
  volume=1,
  edition="Third"}

Other Useful Tools/Topics for LaTeX

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